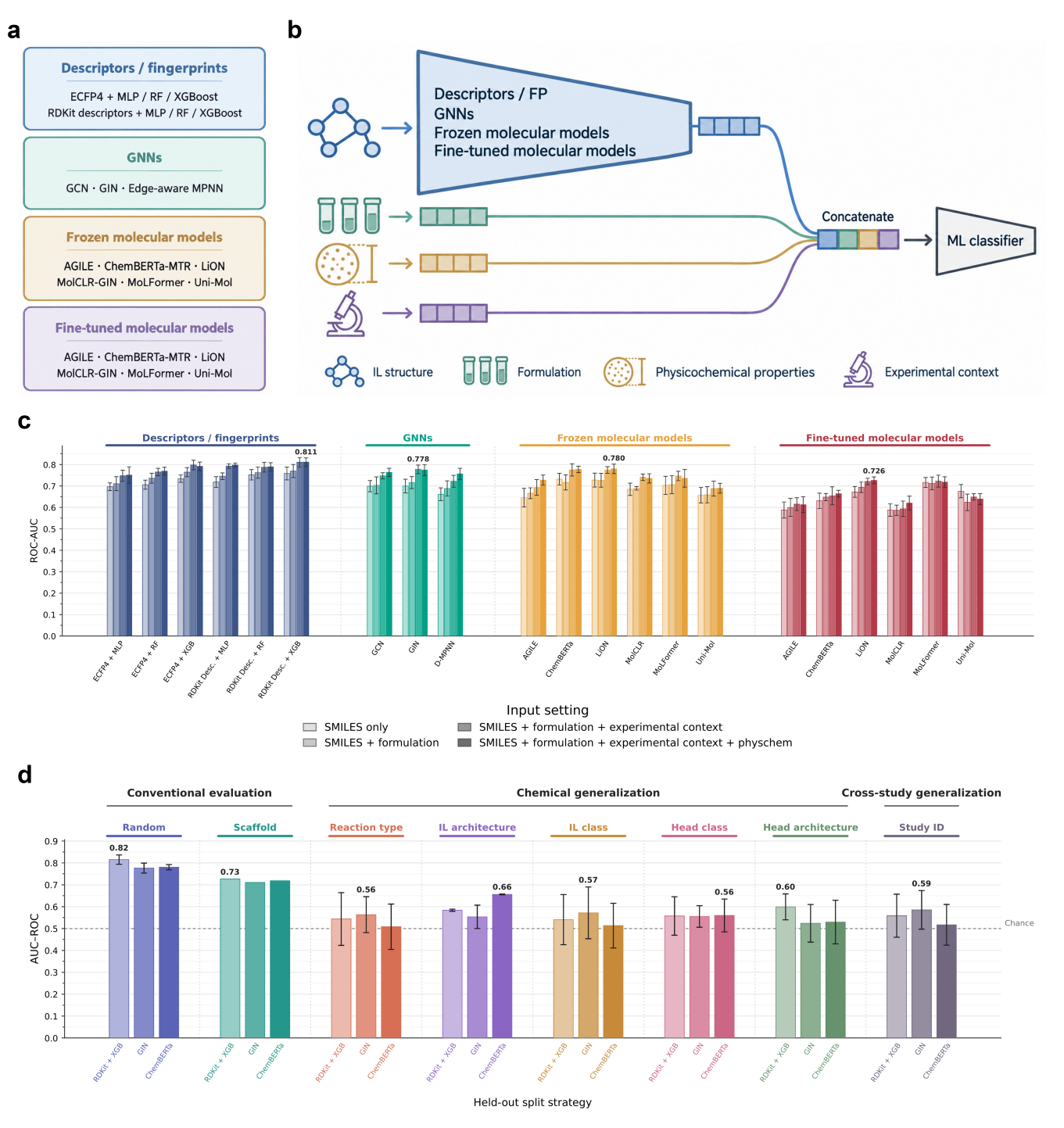

a
Descriptors / fingerprints
ECFP4 + MLP / RF / XGBoost
RDKit descriptors + MLP / RF / XGBoost
GNNs
GCN · GIN · Edge-aware MPNN
Frozen molecular models
AGILE · ChemBERTa-MTR · LiON
MolCLR-GIN · MolFormer · Uni-Mol
Fine-tuned molecular models
AGILE · ChemBERTa-MTR · LiON
MolCLR-GIN · MolFormer · Uni-Mol
b
Descriptors / FP
GNNs
Frozen molecular models
Fine-tuned molecular models
Concatenate
ML classifier
IL structure
Formulation
Physicochemical properties
Experimental context
c
Descriptors / fingerprints
GNNs
Frozen molecular models
Fine-tuned molecular models
ROC-AUC
0.811
0.778
0.780
0.726
ECFP4 + MLP
ECFP4 + RF
ECFP4 + XGB
RDKit Desc. + MLP
RDKit Desc. + RF
RDKit Desc. + XGB
GCN
GIN
D-MPNN
AGILE
ChemBERTa
LiON
MolCLR
MolFormer
Uni-Mol
Input setting
SMILES only
SMILES + formulation
SMILES + formulation + experimental context
SMILES + formulation + experimental context + physchem
d
Conventional evaluation
Chemical generalization
Cross-study generalization
Random
Scaffold
Reaction type
IL architecture
IL class
Head class
Head architecture
Study ID
AUC-ROC
0.82
0.73
0.56
0.66
0.57
0.56
0.60
0.59
Chance
RDKit + XGB
GIN
ChemBERTa
Held-out split strategy